

Unit 4 Guided Notes — Geometry: Angles & Triangles

Name the relationship, write the equation, solve for x . Angles always add up to something.

Standard: NC.8.G.5

This packet has three parts: **Part 1 — Guided Notes**, **Part 2 — Practice Problems**, and **Part 3 — Answer Key**. Fill the blanks on your printed copy first, then click any blank, step, or check button to reveal the answer.

Topics Covered: 4.1 Angle Relationships — Complementary, Supplementary, Vertical, Linear Pair · 4.2 Parallel Lines & Transversals — All Angle Pair Types · 4.3 Triangle Angle Sum & Exterior Angle Theorem (all NC.8.G.5)

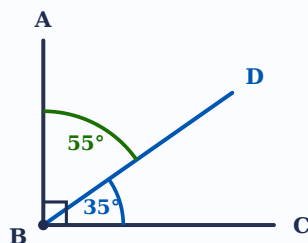
1 Angle Relationships

Fill the definitions and rules on your printed copy, then click to check.

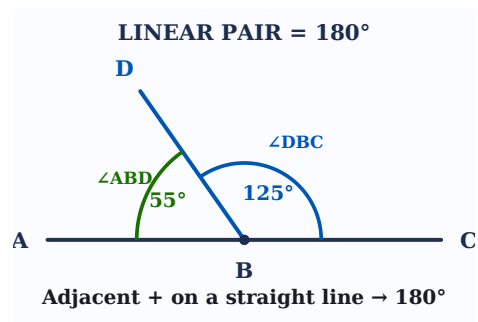
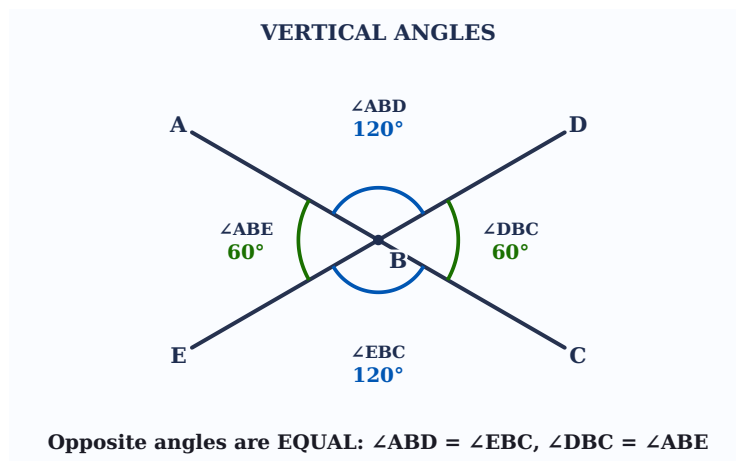
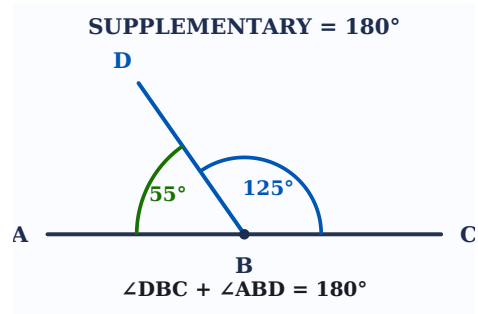
Relationship	Definition	Sum / Rule
Complementary	Two angles that add to °	
Supplementary	Two angles that add to °	
Vertical Angles		
Linear Pair		

Each angle is named with three letters — the **vertex is the middle letter** (e.g. $\angle ABD$ has vertex B). Arcs show which angle each measure belongs to.

COMPLEMENTARY = 90°



$$\angle ABD + \angle DBC = 90^\circ$$



Quick Check

MC Two angles form a linear pair. One measures 115° . What is the other?

✓ A linear pair adds to 180° . $180^\circ - 115^\circ = 65^\circ$.

Match the Relationship to its Rule

MATCH Click a relationship, then its rule.

Relationship

Complementary

Supplementary

Vertical Angles

Linear Pair

Rule

Always EQUAL

Add to 90°

Adjacent angles on a line $\rightarrow 180^\circ$

Add to 180°

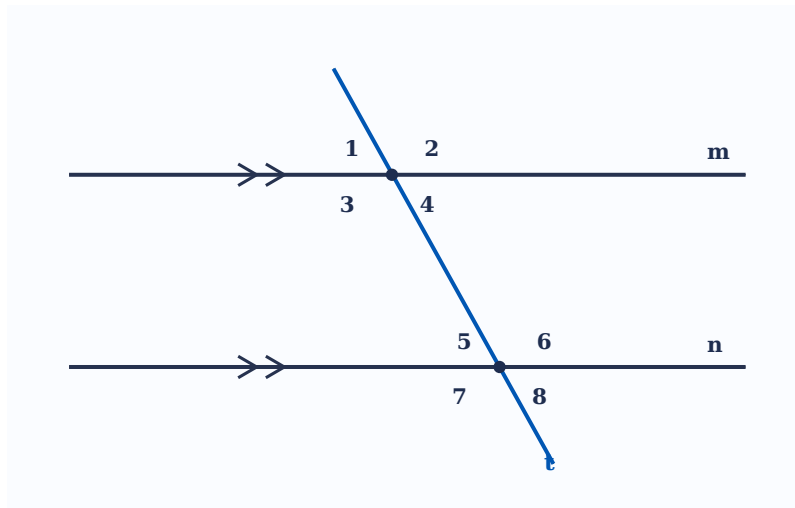
0 of 4 matched

2 Parallel Lines & Transversals

★ REMEMBER:

- Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are CONGRUENT (equal).
- Same-Side Interior (co-interior) and Same-Side Exterior angles are SUPPLEMENTARY (add to 180°).

Two parallel lines ($m \parallel n$) cut by transversal t form 8 angles. Corresponding: (1,5)(2,6)(3,7)(4,8) · Alt. Interior: (3,6)(4,5) · Alt. Exterior: (1,8)(2,7) · Same-Side Interior: (3,5)(4,6)



Fill the Relationship and Rule columns on your printed copy, then click to check.

Angle Pair Type	Relationship	Rule to Use	Example
Corresponding			$68^\circ \rightarrow x = 68^\circ$
Alternate Interior			$101^\circ \rightarrow x = 101^\circ$
Alternate Exterior			$142^\circ \rightarrow x = 142^\circ$

Vertical			$120^\circ \rightarrow x = 120^\circ$
Same-Side Interior (Co-Interior)			$134^\circ \rightarrow x = 46^\circ$
Same-Side Exterior			$77^\circ \rightarrow x = 103^\circ$

Worked Example — Click each step to reveal

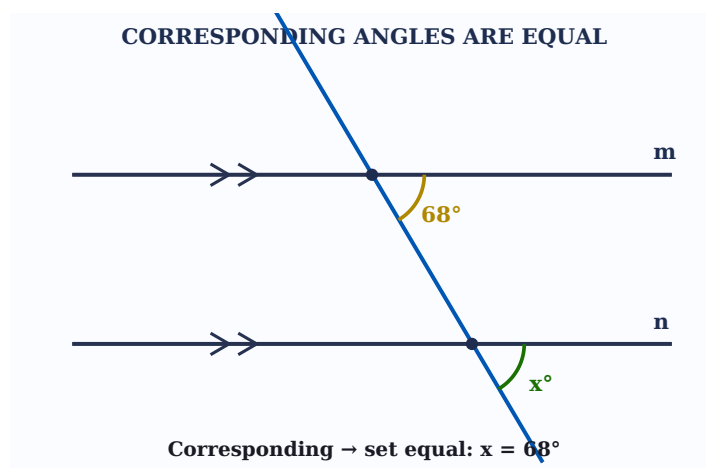
$\angle 1 = 3x + 15$ and $\angle 2 = 5x - 7$ (alternate interior angles)

Alternate interior angles are EQUAL: $3x + 15 = 5x - 7$

Subtract $3x$: $15 = 2x - 7$

Add 7: $22 = 2x$

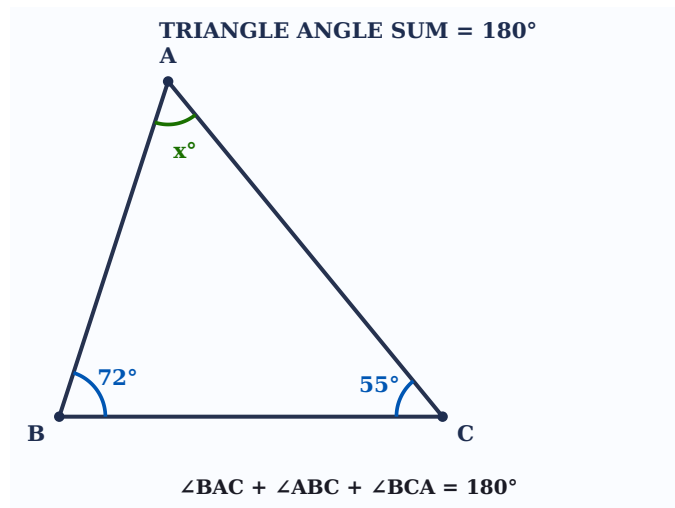
✓ Divide by 2: $x = 11 \rightarrow \text{angle} = 3(11) + 15 = 48^\circ$



3 Triangle Theorems

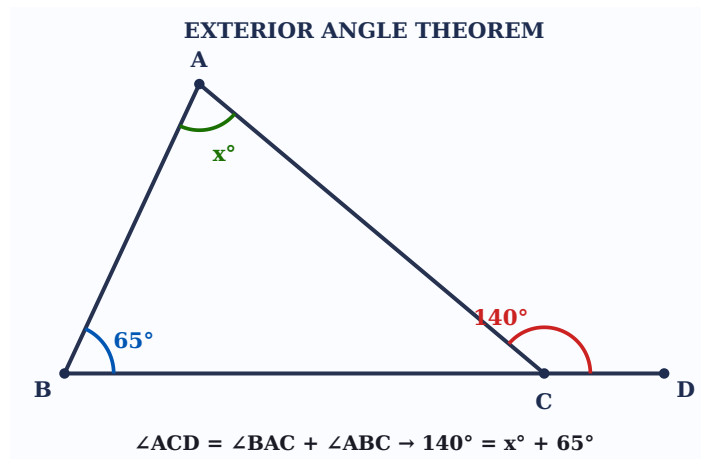
✦ **TRIANGLE ANGLE SUM:** The three interior angles of a triangle always add up to °.

✦ **EXTERIOR ANGLE THEOREM:** An exterior angle equals the of the two non-adjacent (remote) interior angles.



Problem 1 — Triangle Angle Sum: $x + 72 + 55 = 180$

Solve: $x + 127 = 180 \rightarrow x = 53^\circ$



Problem 2 — Exterior Angle Theorem: $x + 65 = 140$

Solve: $x + 65 = 140 \rightarrow x = 75^\circ$

ERROR ANALYSIS **Jordan's Exterior Angle Mistake —** A triangle has interior angles 35° and 70° . Exterior angle = x . Jordan wrote $x = 70 - 35 = 35^\circ$.

Jordan subtracted instead of adding. The exterior angle = SUM of the remote interior angles: $x = 35 + 70 = 105^\circ$.

PART 2 — Practice Problems

Identify the relationship, write the equation, and solve for x . Check on screen.

A Angle Relationships — Solve for x

Key Facts: Complementary = 90° | Supplementary = 180° | Vertical = Equal | Linear Pair = 180° . (Type just the number, e.g. 55.)

1 Supplementary Angles, given 125° . $x = ?$

x =

2 Complementary Angles, given 47° . $x = ?$

x =

3 Vertical Angles, given 72° . $x = ?$

x =

4 Linear Pair, given 138° . $x = ?$

x =

5 Complementary Angles, given 35° . $x = ?$

x =

6 Supplementary Angles, given 110° . $x = ?$

x =

7 Vertical Angles, given 63° . $x = ?$

x =

8 Linear Pair, given 157° . $x = ?$

x =

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9 Complementary Angles, given 52° . $x = ?$

$x =$

10 Supplementary Angles, given 95° . $x = ?$

$x =$

B Parallel Lines & Transversals

Rules: Corresponding = Equal | Alternate Interior = Equal | Alternate Exterior = Equal | Co-Interior (Same-Side) = 180° .

1 Corresponding, given 65° . $x = ?$

$x =$

2 Alternate Interior, given 110° . $x = ?$

$x =$

3 Alternate Exterior, given 75° . $x = ?$

$x =$

4 Corresponding, given 48° . $x = ?$

$x =$

5 Alternate Interior, given 130° . $x = ?$

$x =$

6 Co-Interior, given 55° . $x = ?$

$x =$

7 Alternate Exterior, given 92° . $x = ?$

$x =$

8 Co-Interior, given 140° . $x = ?$

$x =$

C Triangle Theorems

Interior Sum: All 3 angles = 180° . **Exterior Angle:** = sum of the 2 remote interior angles.

1 Angle Sum, given 65° , 80° . $x = ?$

$x =$

2 Angle Sum, given 40° , 90° . $x = ?$

$x =$

3 Angle Sum, given 53° , 55° . $x = ?$

$x =$

4 Angle Sum, given 30° , 90° . $x = ?$

$x =$

5 Exterior Angle Theorem, remotes 45° , 75° . Exterior $x = ?$

$x =$

6 Exterior Angle Theorem, remote 55° , exterior = 135° . $x = ?$

$x =$

7 Exterior Angle Theorem, remote 46° , exterior = 118° . $x = ?$

$x =$

8 Angle Sum, given 50° , 50° . $x = ?$

$x =$

★ PART 3 — Answer Key ★

Do not distribute to students until after the lesson.

★ Answers

Guided Notes

4.1: Complementary $\rightarrow 90^\circ$ ($a + b = 90^\circ$); Supplementary $\rightarrow 180^\circ$ ($a + b = 180^\circ$); Vertical = opposite angles when lines cross, always EQUAL; Linear Pair = adjacent angles on a straight line, add to 180° . 4.2: Corresponding/Alt. Interior/Alt. Exterior/Vertical = CONGRUENT (set equal); Same-Side Interior & Exterior = SUPPLEMENTARY (add to 180°). Worked: $3x + 15 = 5x - 7 \rightarrow x = 11 \rightarrow$ angle = 48° . 4.3: Angle Sum = 180° ; Exterior = SUM of remote interiors. Problem 1: $x = 53^\circ$. Problem 2: $x = 75^\circ$. Error analysis: $x = 35 + 70 = 105^\circ$.

Part A — Angle Relationships

1) 55° 2) 43° 3) 72° 4) 42° 5) 55° 6) 70° 7) 63° 8) 23° 9) 38° 10) 85° .

Part B — Parallel Lines

1) 65° 2) 110° 3) 75° 4) 48° 5) 130° 6) 125° ($180 - 55$) 7) 92° 8) 40° ($180 - 140$).

Part C — Triangle Theorems

1) 35° 2) 50° 3) 72° 4) 60° 5) Exterior = $45 + 75 = 120^\circ$ 6) $135 = 55 + x \rightarrow 80^\circ$ 7) $118 = 46 + x \rightarrow 72^\circ$ 8) 80° .

← Back to Unit 4